

SALWA MHEMED

+216 92789497 salwa.mhemed05@gmail.com [LinkedIn](#) [GitHub](#) salwa-mhemed.netlify.app

Experience

Data Science Intern

April 2025 - September 2025

Unisphere GmbH

Germany - On site

- Built a Python pipeline to generate simulation inputs for the Trajeanie drone performance tool from real telemetry and flight plan data.
- Analyzed flights from multiple drone data providers including Skyflow, ANWB, and RigiTech; conducted sensitivity studies; and produced visual/statistical reports to support model evaluation.
- Identified biases in flight duration and energy consumption estimates and provided recommendations to refine the model.

Machine Learning Intern

June 2024 - March 2025

Dydon AI (June 2024 - August 2024) / Part-time (October 2024 - March 2025)

Switzerland - Remote

- Conducted research on NLP models for financial document processing, exploring different architectures and fine-tuning approaches to enhance key-value extraction accuracy.
- Developed an AI pipeline to automate the extraction and structuring of financial data from PDF documents using PyMuPDF, Transformers (Hugging Face), and Llama, improving processing efficiency.

Web Development Intern

June 2023 - July 2023

Aures Group

Tunisia - Hybrid

- Developed user-friendly interfaces using React.js, with a responsive design for seamless cross-platform navigation.
- Implemented a Node.js back-end API with MongoDB for secure data handling and real-time updates to product inventory and orders.

Projects

End-to-End Churn Prediction System using MLflow, Docker, and CI/CD

[Source Code](#)

- Developed and trained an XGBoost-based churn prediction model with feature engineering and class imbalance handling, achieving 72.5% test accuracy, 0.838 ROC-AUC, and 0.615 F1-score, with strong recall for churn detection.
- Built a full end-to-end ML pipeline including data preprocessing, feature engineering (7,000+ features), model evaluation, and experiment tracking using MLflow and Great Expectations.
- Deployed the model using FastAPI with a Gradio interface and containerized the application using Docker for reproducibility and scalability, while implementing CI/CD pipelines with GitHub Actions to automate builds and streamline deployment and updates.

AI-Based Series Analysis System using LLMs and NLP

[Source Code](#)

- Developed a theme classification algorithm using Hugging Face zero-shot learning in Python.
- Built an NER system with spaCy to extract character names and visualize relationships using NetworkX and PyVis.
- Created a character chatbot with a fine-tuned LLaMA model and integrated Gradio for interactive visualization.

Multi-PDF Conversational AI Application using Retrieval-Augmented Generation

[Source Code](#)

- Implemented a retrieval-augmented generation (RAG) pipeline using LangChain, FAISS, and Hugging Face embeddings to process and query PDF documents with context-aware responses.
- Built an interactive Streamlit-based chatbot enabling users to upload PDFs and interact conversationally using an LLM with memory and vector search.

AI-Based Tennis Match Analysis System using Computer Vision

[Source Code](#)

- Developed, fine-tuned, and trained YOLOv8 models for detecting and recognizing players, tennis balls, and court keypoints, and drew bounding boxes to visualize detections on video frames.
- Created a mini-court visualization using OpenCV to represent players and game dynamics, mapping player and ball movements on a simplified court model.
- Implemented shot detection and speed analysis by calculating ball and player speeds, and visualized key statistics such as shot count and player movement speeds. Annotated match videos with court keypoints and player statistics, providing a detailed performance summary.

Skills

Programming: Python, C/C++, JavaScript, MATLAB, HTML/CSS

Machine Learning & AI: TensorFlow, Scikit-learn, LangChain, Hugging Face Transformers, LLMs, RAG, YOLOv8

Data, Visualization & Tools: NumPy, Pandas, Matplotlib, PostgreSQL, Git, OpenCV, NetworkX, PyVis, Streamlit, Gradio, FastAPI, Docker, CI/CD

Education

National Institute of Applied Sciences and Technology (INSAT)

2020 - 2025

Industrial IT and Automation Engineering Degree

Certifications

Machine Learning Specialization by Stanford University & DeepLearning.AI

Mars 2024

Languages

• **Arabic:** Native **English:** Advanced **French:** Advanced **German:** Beginner